

PHASE 1 · BUILD SET



Kids' Treehouse

Freestanding post platform around a single tree · 8'×8' · 15° slope · NJ

Tree

17.5" dia (55" circ) · likely hickory/ash

Platform

8' × 8' level deck

6 posts on concrete footings

trunk through a 26" hole (≈4" gap)

Heights

uphill edge ≈28" · downhill ≈54"

grade drop ≈26" across the run

Structure

2× doubled 2×8 beams

2×8 joists @16" o.c.

5/4×6 PT decking

36" railing, <4" baluster gaps

Rough cost

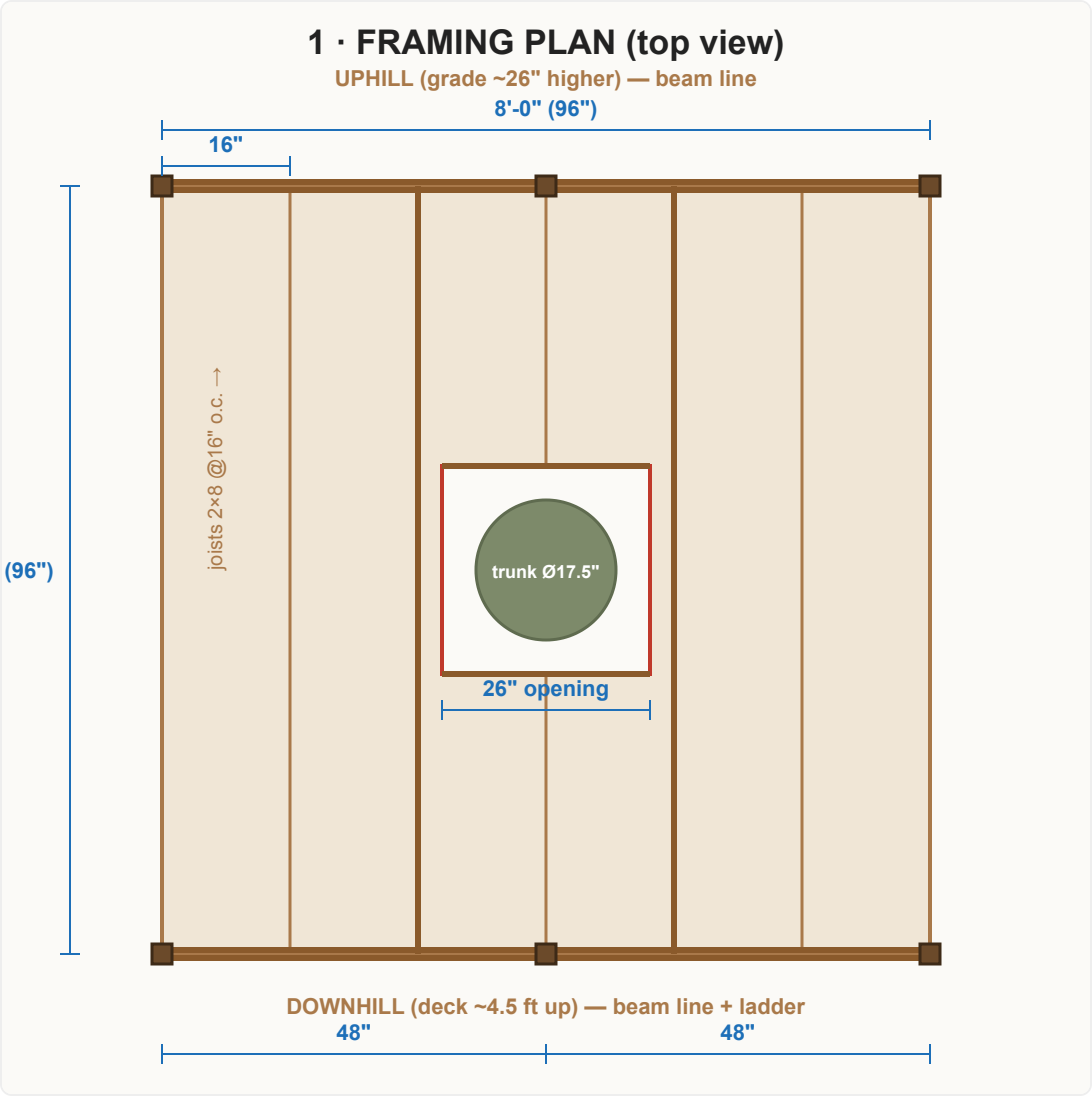
\$700–1,100

Interactive 3D

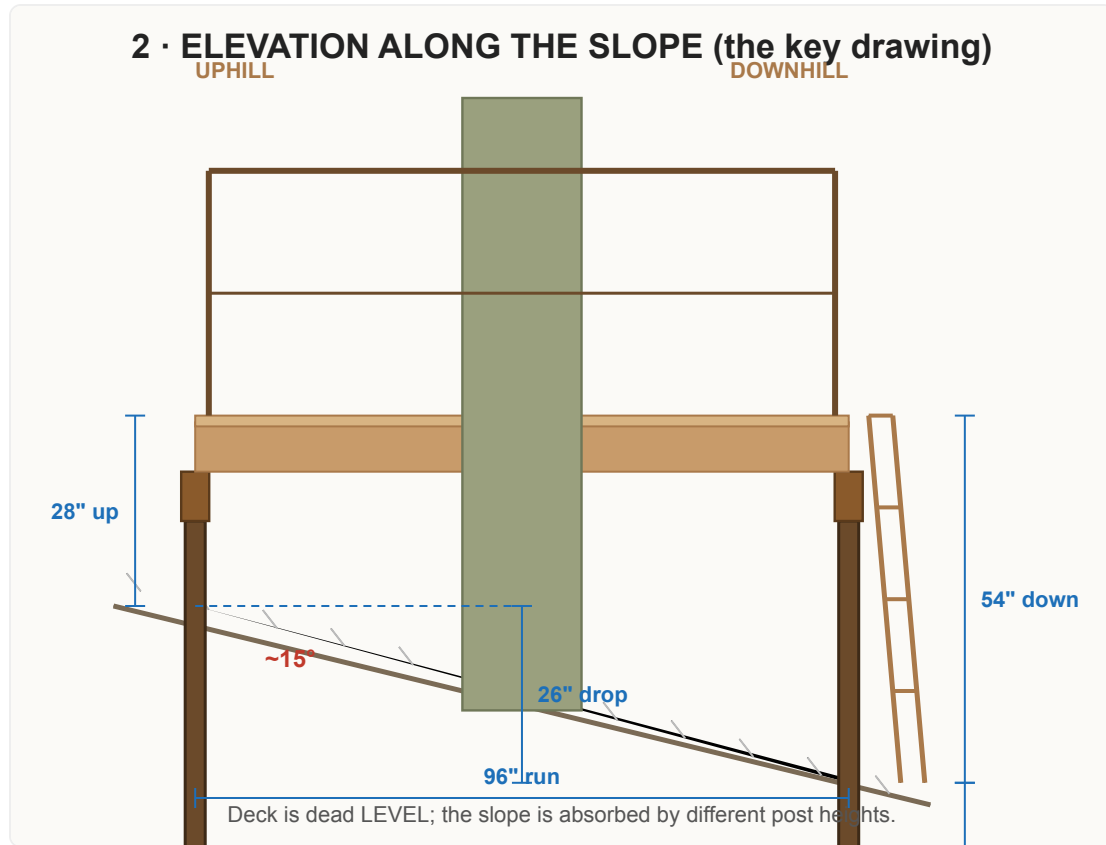
[open plans/model3d.html](#)

Drawings are dimensioned planning set — verify on site. Pressure-treated lumber throughout.

1 · Framing plan

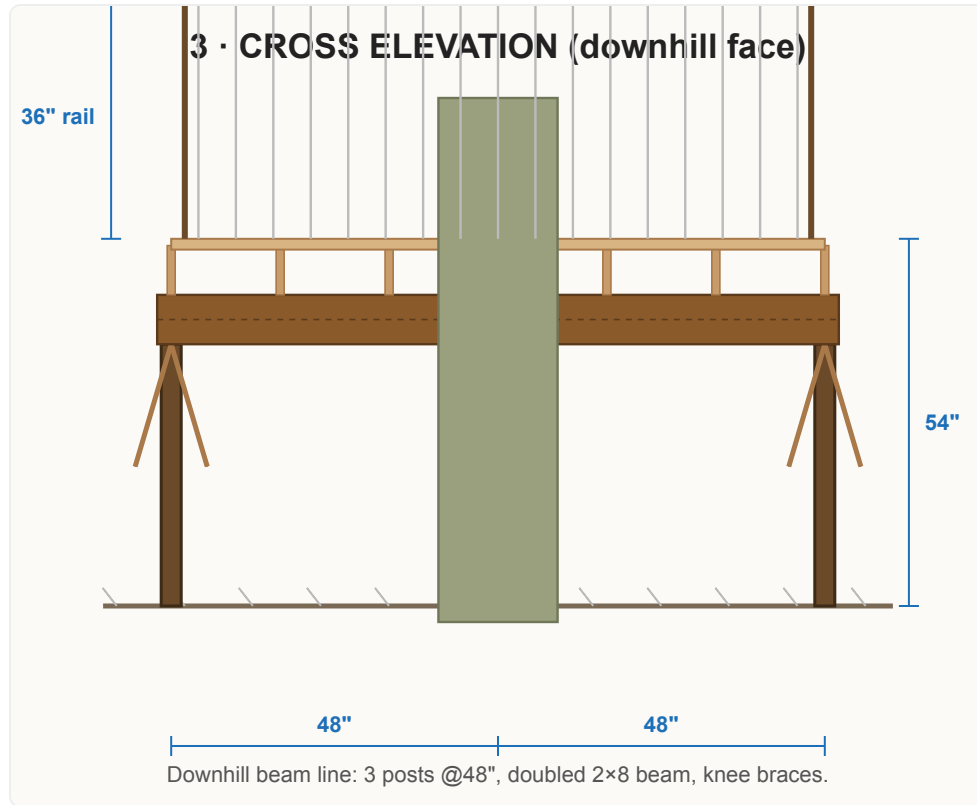


2 · Elevation along the slope (key drawing)

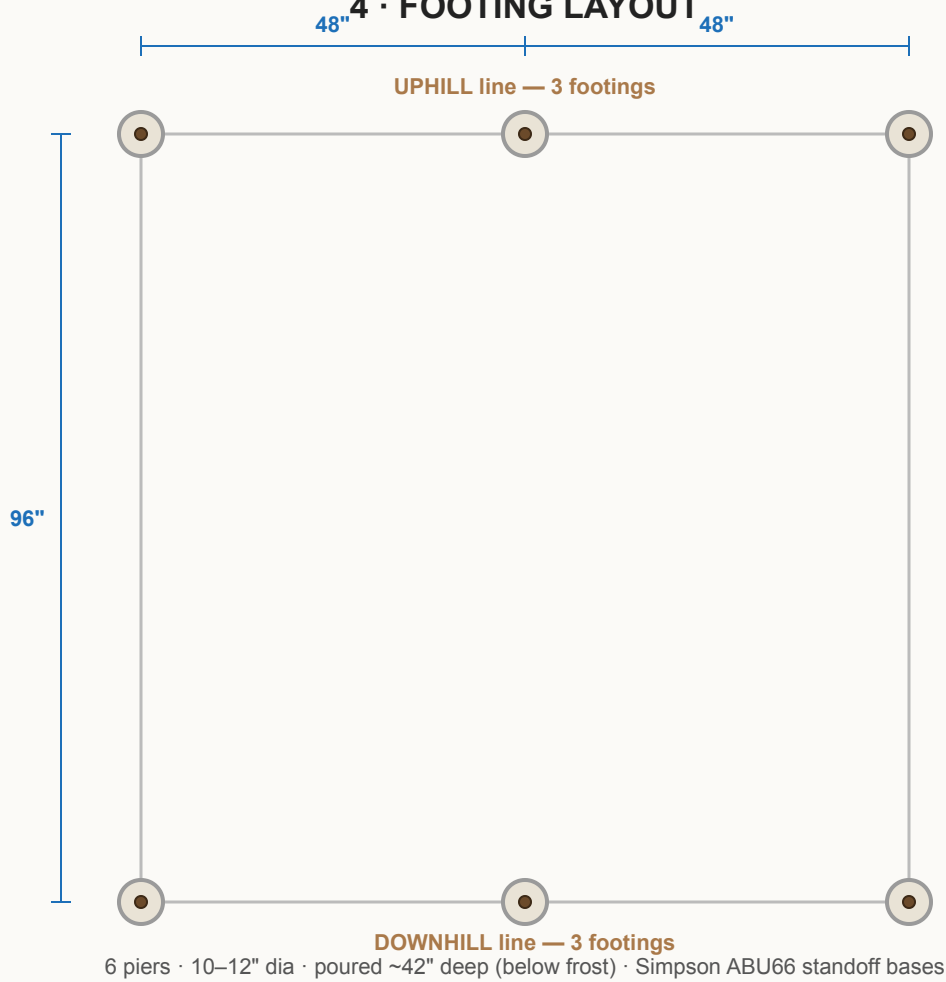


The deck is dead level; the slope is absorbed entirely by different post heights.

3 · Cross elevation · 4 · Footings

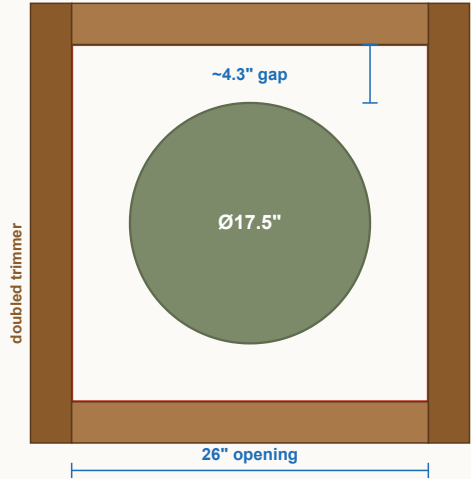


4 · FOOTING LAYOUT



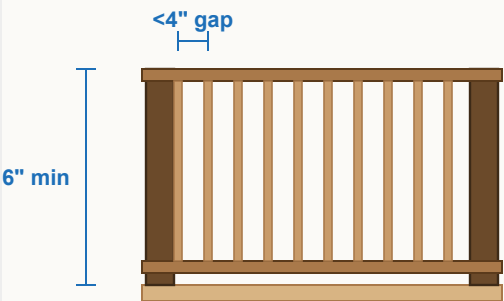
5 · Tree-opening detail · 6 · Railing detail

5 · TREE-OPENING DETAIL



~4" clear gap all around for growth + sway. Framing never touches bark.

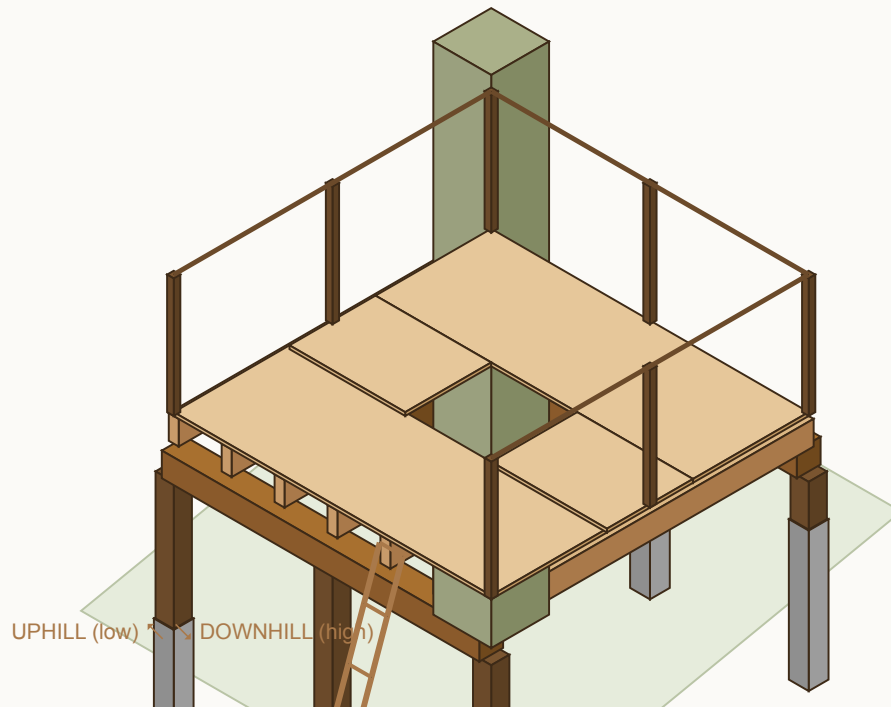
6 · RAILING DETAIL (kid safety)



Vertical 2×2 balusters; no gap > 4". 4×4 posts bolted to rim joist.

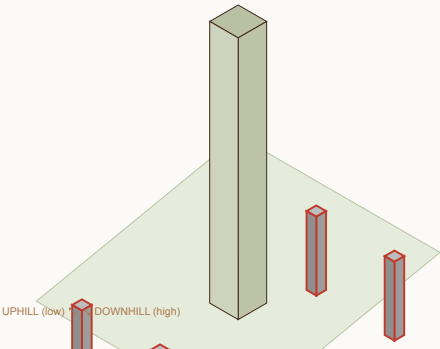
7 · Isometric assembly (3D)

7 · ISOMETRIC ASSEMBLY (3D)



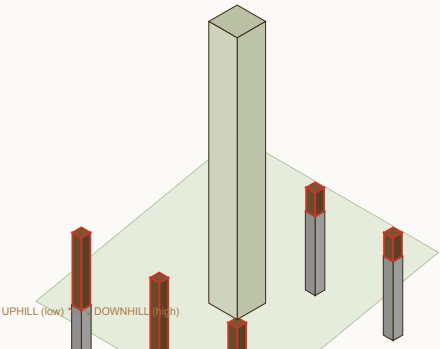
Build sequence (3D, step by step)

01 · Pour 6 footings



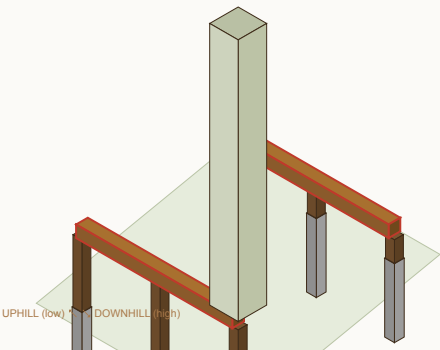
1 · Pour 6 footings

02 · Set 6 posts, cut level



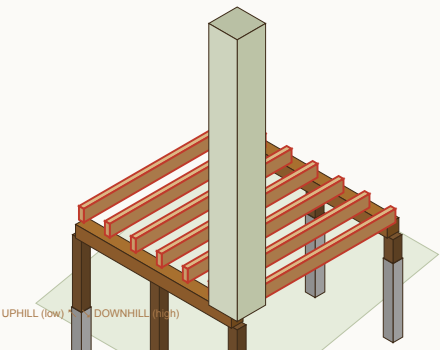
2 · Set posts, cut level

03 · Bolt on 2 doubled beams



3 · Doubled beams

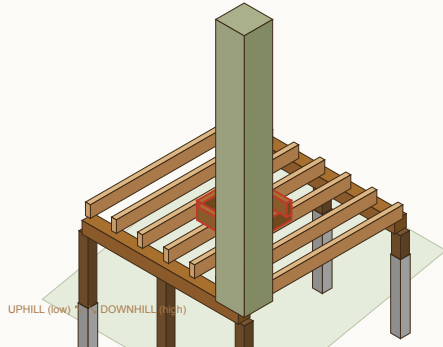
04 · Lay joists @16" o.c.



4 · Joists @16" o.c.

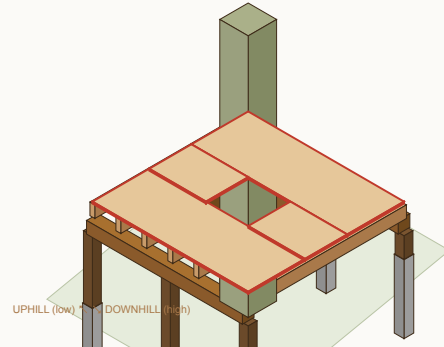
Build sequence (cont.)

05 · Frame tree opening + trunk



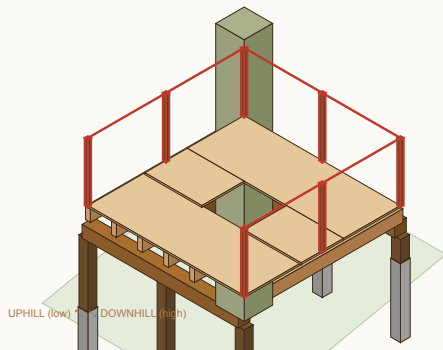
5 · Frame tree opening

06 · Screw down decking



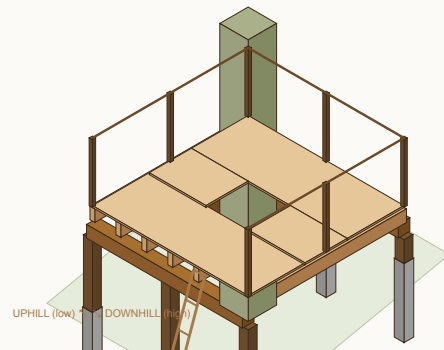
6 · Decking

07 · Build the railing



7 · Railing

08 · Add ladder / steps



8 · Ladder & finish

Bill of materials (8×8, Phase 1)

Item	Spec	Qty	Notes
Posts	6×6 PT, 8 ft	6	Cut to a level line
Beams	2×8 PT, 8 ft	4	Two doubled beams
Joists	2×8 PT, 8 ft	11	+ doubling at opening
Knee braces	4×4 PT, 8 ft	2	Cut to 45° braces
Decking	5/4×6 PT, 8 ft	20	64 sq ft + waste
Rail posts	4×4 PT, 8 ft	4	~48" each
Rails	2×4 PT, 8 ft	8	Top + bottom
Balusters	2×2 PT, 8 ft	~12	<4" gaps
Concrete	60 lb bags	~15	6 piers ~42" deep
Tube forms	10–12" Sonotube	6	
Post bases	Simpson ABU66	6	Standoff, in wet concrete
Joist hangers	Simpson LUS28	~24	+ hanger nails
Carriage bolts	½" × 8"	~20	Beams to posts
Screws	structural + deck	—	GRK/SDWS + coated deck

Rough total \$700–1,100 depending on PT prices. Verify post lengths on site after footings cure.